AGE 310: Instrumentation and Measurements

Dr. Kadeghe Fue, Department of Agricultural Engineering, SUA

Morogoro, Tanzania

Abstract

Lecture 9 investigates the standardized rules (*protocols*) necessary for data exchange between electronic components. We compare basic transmission methods (Parallel vs. Serial) and analyze the fundamental digital communication protocols (UART, I2C, SPI) used for short-range sensor integration. Critical focus is placed on the agricultural industry standards: CAN Bus and ISOBUS, which enable interoperability between tractors and implements.

Lecture 9: Data Communication Protocols

9.1 Transmission Fundamentals

9.1.1 Parallel vs. Serial Transmission

Data Transmission is the process of sending binary data (bits) from one point to another.

- **Parallel Transmission:** Multiple bits are sent simultaneously over multiple dedicated lines (e.g., 8 lines for an 8-bit byte).
  - **Pros:** Very fast over short distances.
  - **Cons:** Requires bulky cables; prone to *crosstalk* and timing skew over long distances, making it impractical for field use.
- **Serial Transmission:** Bits are sent sequentially, one after the other, over a single data line.
  - **Pros:** Simple, minimal wiring (cheap), highly reliable over long distances (due to lower capacitance).
  - **Cons:** Slower than parallel at the same clock speed.

Serial is the standard for almost all agricultural field instrumentation.

9.1.2 Synchronous vs. Asynchronous Communication

- **Synchronous (Clocked):** Uses a separate line to transmit a clock signal (SCK or SCL) alongside the data. The receiver uses the clock edge to precisely synchronize the reading of each bit.
  - **Examples:** I2C, SPI.
- **Asynchronous (Un-clocked):** No clock line is used. Both the transmitter and receiver must agree on a **Baud Rate** (speed) beforehand. The receiver uses start/stop bits to frame the data.
  - **Example:** UART.

9.2 Basic Microcontroller Protocols

9.2.1 UART (Universal Asynchronous Receiver-Transmitter)

UART is the most fundamental serial protocol, typically used for debug output or simple data streaming.

- **Wires:** 2 (TX - Transmit, RX - Receive).
- **Topology:** Point-to-Point (one transmitter talks to one receiver).
- **Data Format:** Data is sent in packets, preceded by a start bit and followed by stop bits (framing).
- **Ag Use:** Receiving NMEA sentences (data packets) from standard GPS modules, communicating with GSM modems for remote data transmission.

9.2.2 I2C (Inter-Integrated Circuit)

I2C (or Two-Wire Interface, TWI) is designed for communicating between components on a single circuit board or short cables.

- **Wires:** 2 (SDA - Data, SCL - Clock).
- **Topology:** Multi-Master/Multi-Slave. Multiple devices share the same bus.
- **Addressing:** Each slave sensor has a unique 7-bit address (e.g., $0\text{x}40, 0\text{x}68$). The master selects the slave by sending its address first.
- **Ag Use:** Connecting multiple sensors (temperature, barometric pressure, humidity) to a single microcontroller using only two wires, common in compact weather stations.

9.2.3 SPI (Serial Peripheral Interface)

SPI is a high-speed synchronous protocol popular for interfacing with memory devices and displays.

- **Wires:** 4 (SCK - Clock, MOSI - Master Out Slave In, MISO - Master In Slave Out, SS/CS - Slave Select/Chip Select).
- **Topology:** Master/Slave.
- **Speed:** Full-duplex (sending and receiving simultaneously), making it the fastest of the three.
- **Ag Use:** Reading and writing data to SD card modules for data logging, driving graphical user interfaces (GUIs) on field monitors.

9.3 The Agricultural Standard: CAN Bus

9.3.1 Controller Area Network (CAN)

CAN Bus was developed for automotive applications but became the backbone of heavy-duty vehicle electronics (J1939 standard) and, by extension, agricultural machinery.

- **Robustness:** CAN uses a **Differential Signal** over two wires (CAN_High and CAN_Low). These wires carry inverted signals. Noise affects both wires equally, but the receiver only looks at the *difference* between them, resulting in extremely high Common Mode Rejection (CMRR).
- **Data Priority:** Messages are prioritized by their *ID*. High-priority messages (e.g., Engine STOP) always interrupt lower-priority messages (e.g., GPS data).
- **Topology:** Multi-Master, Bus Topology. All nodes can transmit, making it robust against single-point failures.

9.3.2 ISOBUS (ISO 11783)

ISOBUS is the specific adaptation of the CAN Bus protocol for agricultural machinery, established under the ISO 11783 standard.

- **Goal:** To enable "Plug-and-Play" operation. Any compliant tractor (*e.g.*, John Deere) can communicate with any compliant implement (*e.g.*, Kuhn Sprayer) using a standard cable and one in-cab display (*Virtual Terminal*).
- **Components:** Defines standard messages for functions like implement control, rate control, task control, and GPS data exchange.
- **Ag Impact:** Reduces clutter and cost, fostering interoperability among different manufacturers for precision applications.

9.4 Wireless Telemetry and IoT

9.4.1 Wireless Protocols

- **Wi-Fi (IEEE 802.11):** High data rate, medium range (~ 100 m), high power consumption. Ideal for greenhouses or farm offices with constant power.
- **Bluetooth/BLE:** Low data rate, short range (~ 10 m), very low power. Used for configuration and monitoring via smartphone apps in the field.
- **GSM/4G (Cellular):** Long range, high data rate, high power consumption. Used for remote data transmission where public cellular networks exist (common for IoT in rural Tanzania).

9.4.2 LoRaWAN (Long Range Wide Area Network)

LoRaWAN is specifically designed for IoT in vast rural areas like agricultural fields.

- **Range:** Extremely long range (up to 15 km or more line-of-sight).
- **Power:** Ultra-low power. Battery-powered sensors can last years.
- **Data Rate:** Very low data rate (best for small packets like temperature or moisture readings every hour).
- **Architecture:** Devices (*Nodes*) communicate with a *Gateway*, which connects to the *Cloud*. This is highly scalable for large farms.

Calculation Example 1: Data Rate Comparison

A sensor sends 100 bytes of data every second. Compare the time required to transmit this data using a fast SPI link running at 1 MHz vs. a typical UART link running at 9600 Baud. (Assume 1 Byte $= 8$ bits for SPI and 10 bits for UART due to start/stop bits).

Solution:

- **Total Bits per Second:** 100 Bytes/s $\times$ 8 bits/Byte $= 800$ bits/s (For SPI data payload)
- **SPI Transmission Time (T_{SPI}):** Baud Rate $= 1$ MHz $= 1,000,000$ bits/s

$$T_{\text{SPI}} = \frac{800 \text{ bits}}{1,000,000 \text{ bits/s}} = 0.0008 \text{ s} = 0.8 \text{ µs}$$

- **UART Transmission Time (T_{UART}):** Baud Rate $= 9600$ bits/s. (Total bits including framing $= 100 \times 10 = 1000$ bits).

$$T_{\text{UART}} = \frac{1000 \text{ bits}}{9600 \text{ bits/s}} \approx 0.104 \text{ s} = 104,000 \text{ µs}$$

The SPI link is approximately $130,000$ times faster than the UART link, demonstrating why protocols are selected based on data volume and speed needs.

9.5 GPS Data and Standard Formats

The Global Positioning System (GPS) is fundamental to precision agriculture. The most common format for transmitting GPS data is the NMEA 0183 standard.

NMEA Sentence: A single line of text data that begins with $, e.g.,

```
$GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47
```

Each field is separated by a comma (delimiter). Microcontrollers use UART to read these strings and then parse them to extract values like time, latitude, and altitude.

Calculation Example 2: Decoding NMEA Data

Decode the critical fields in the NMEA sentence given above:

```
$GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47
```

Solution (Field Indexing):

- **Field 2 (Time):** 123519 (UTC time: $12:35:19$)
- **Field 3/4 (Latitude):** $4807.038,$ N (48 degrees, 07.038 minutes North)
- **Field 5/6 (Longitude):** $01131.000,$ E (11 degrees, 31.000 minutes East)
- **Field 7 (Fix Quality):** 1 (Valid GPS fix)
- **Field 10 (Altitude):** 545.4 (meters above MSL)

Microcontrollers run algorithms to perform this parsing and conversion.

9.6 Summary of Key Terms (Swahili/Context)

- **Serial/Parallel:** Mfululizo/Sambamba (Sequence vs. Simultaneous).
- **Baud Rate:** Kasi ya biti (Transmission speed).
- **UART:** UART (Universal Asynchronous Receiver-Transmitter).
- **I2C/SPI:** Aina za Serial zinazoongozwa na Clock.
- **CAN Bus:** CAN Bus (Mfumo wa Udhibiti wa Eneo - standard ya matrekta).
- **ISOBUS:** ISOBUS (Standard ya Plug-and-Play ya Kilimo).
- **LoRaWAN:** Mtandao wa Masafa Marefu (Long-range, low-power network).
- **Telemetry:** Mawasiliano ya Mbali (Remote data transmission).
- **NMEA:** Lugha ya GPS (Standard data format for GPS).

Lecture 9: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Define *Serial Transmission* and *Parallel Transmission*. Explain why all agricultural IoT systems prefer Serial communication for sensors deployed in the field.
2. Differentiate between *Synchronous* and *Asynchronous* serial communication protocols. Give one example of each.
3. Explain the role of the 7-bit address in the I2C protocol. Why does the SPI protocol not rely on addressing?
4. Define the term *Differential Signal* as used by CAN Bus. Why does this technique provide superior noise immunity in a vibrating tractor environment?
5. What is ISOBUS (ISO 11783)? State its primary benefit for Tanzanian farmers who use implements from different global manufacturers.

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **SPI vs. I2C Selection:** An agricultural robot requires data from 4 high-resolution image sensors (fast data transfer) and 10 low-speed temperature sensors.
   - Recommend the best serial protocol (I2C or SPI) for the 4 image sensors and justify your choice.
   - Recommend the best protocol for the 10 temperature sensors and explain why.
7. **UART Framing Error:** UART communication is set to 115200 Baud. If the transmitting microcontroller is mistakenly configured to run 1% slower than the receiver, calculate the cumulative timing error (in ms) after 10 complete bytes (assuming 10 bits per byte).
8. **LoRaWAN Data Rate:** A LoRaWAN sensor transmits data packets totaling 50 bytes every hour. Assuming 1 day $= 86,400$ seconds, calculate the average data rate in bits per second (bps) used by this sensor over one day. (1 Byte $= 8$ bits).
9. **GPS NMEA Parsing:** Decode the critical fields from the following NMEA sentence to determine the location and altitude (in meters):

   ```
   $GPGGA,092305.00,0648.7891,S,03912.4567,E,1,08,0.9,150.2,M,45.6,M,,*42
   ```

10. **CAN Bus Error Detection:** Explain the *Cyclic Redundancy Check (CRC)* mechanism used in CAN Bus messages. What is the consequence if the CRC calculated by the receiving ECU does not match the CRC transmitted by the source sensor?
11. **Bus Contention (I2C):** Describe what happens in the I2C protocol if two different sensor nodes simultaneously attempt to communicate with the master without having been addressed first. How does the protocol handle this *bus contention*?
12. **Telemetry Selection:** A large-scale commercial farm near Morogoro needs to monitor soil moisture data from 50 fixed sensors spread across 5 km^2. Would you recommend Wi-Fi or LoRaWAN for the network layer? Justify your choice based on range, power consumption, and cost.

13. **CAN Bus Noise Immunity Calculation:** The CAN_H wire carries a 2.5 V signal with a 1 V noise spike, reaching 3.5 V. The CAN_L wire carries 1.5 V with the same 1 V spike, reaching 2.5 V. What differential voltage does the receiver detect, and why is the 1 V noise spike ignored?

14. **SPI Chip Select (CS):** Explain the function of the Chip Select (CS) line in the SPI protocol. How does this line allow the Master to distinguish between multiple Slave devices connected to the same MOSI and MISO lines?

15. **Protocol Layers:** Relate the following three components of a modern precision farming system to the 3-Layer IoT Architecture (Perception, Network, Application):

    - Soil pH Sensor
    - GSM Modem
    - Farm Management Dashboard on a laptop